

Generalized Hierarchical Bayesian Segmentation with Irregular Designs, Multi-Sequence Hierarchies, and Grouped/Latent-Group Designs

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ABSTRACT

BayesBreak is a generalized hierarchical Bayesian method for offline multiple-changepoint segmentation of ordered data. For a regular exponential-family observation model with a proper segmentwise conjugate prior and finite normalizing constants, cumulative sufficient statistics determine the marginal likelihood and posterior moments of each candidate segment. Dynamic-programming recursions then compute the posterior distribution over the number and locations of changepoints, while a separate max-sum recursion with backtracking returns the joint maximum-a-posteriori partition. The partition calculation is therefore common across supported observation families once their segment marginal likelihoods are supplied.

The method extends this construction to irregular design points, multiple conditionally independent sequences with common changepoints, known groups, and a finite latent-group segmentation criterion. Executed studies include Gaussian, Poisson, Binomial, and Beta-valued examples; a repeated Gaussian boundary-calibration study with ECE approximately 0.010 and Brier score approximately 0.011; a two-group latent-allocation study with 96% hard assignment accuracy for its stated design; a finite-range runtime study from 0.0455 seconds at $n = 50$ to 1.0585 seconds at $n = 800$ for $k_{\max} = 20$; and four descriptive real-data applications. These are real executed results, but the application studies lack independent reference changepoints and therefore do not establish external accuracy.

The mathematical core and principal hierarchical extensions are established under explicit assumptions. Remaining work concerns implementation corrections, completion of unresolved tests, controlled recomputation of two excluded historical calculations, broader repeated validation, and final release engineering. This document summarizes the scientific position, current evidence, implementation sequence, resource requirements, and decision criteria without introducing claims beyond the technical book and journal paper.

Keywords Bayesian changepoint analysis, multiple changepoints, product partition models, exponential families, conjugate priors, hierarchical Bayesian models, dynamic programming, irregular designs, multiple sequences, latent groups

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1 Research objective and statistical scope

The research problem is to infer an unknown ordered partition of observations into contiguous regimes while retaining posterior uncertainty about the number and locations of changepoints. The method must support observation models beyond the Gaussian case, accommodate nonuniform design coordinates, and represent dependence among several related sequences through common, group-specific, or latent-group segmentation structures.

Scientific contribution

BayesBreak is a generalized hierarchical Bayesian segmentation method. Its generality comes from combining observation-model-specific segment marginal likelihoods with a common posterior calculation over contiguous partitions. Its hierarchical structure covers irregular designs, multiple sequences, known groups, and finite latent-group allocations under the assumptions stated in the technical paper.

The method belongs to the literature on product-partition models, Bayesian multiple-changepoint analysis, conjugate exponential-family inference, dynamic programming, hierarchical Bayesian modeling, and finite latent-allocation optimization (Barry & Hartigan, 1992, 1993; Diaconis & Ylvisaker, 1979; Fearnhead, 2006; Hutter, 2007). BayesBreak does not claim that conjugate integration or changepoint dynamic programming is new in isolation. The contribution is the generalized statistical formulation and the set of hierarchical extensions developed and evaluated within one method.

1.1 What the generality claim means

For a candidate segment $B_{ij} = \{i, \dots, j\}$, let

$$M_{ij} = \int p(y_{i:j} \mid \theta_{ij}, d_{i:j}) p(\theta_{ij}) d\theta_{ij}$$

be its marginal likelihood, where $d_{i:j}$ denotes family-specific descriptors such as exposures or trial counts. The dynamic program needs the finite array $\{M_{ij}\}$ and the local prior factors for admissible segments and boundaries. It does not require the same sampling distribution for every application.

Exact analytic segment integration is available for the supported regular exponential-family models with proper conjugate priors and finite normalizing constants. When M_{ij} is supplied by quadrature, Laplace approximation, variational approximation, expectation propagation, or another deterministic calculation, the subsequent partition recursion is exact for the supplied score array, but the resulting posterior is an approximation to the intended model unless a segmentwise error bound is established.

1.2 Primary outputs

The method returns distinct inferential quantities:

- the marginal likelihood conditional on the segment count and the posterior distribution of that count;
- posterior changepoint probabilities and ordered-location marginals;
- posterior moments of the piecewise-constant signal, including partition-averaged curves; and
- the joint MAP partition from a separate max-sum recursion with backtracking.

Marginal changepoint modes are not substituted for the joint MAP partition. This distinction is both statistical and computational: marginalization and joint maximization solve different problems.

2 Method and hierarchical extensions

2.1 Segment calculations and partition inference

For a partition $\tau = (0 = t_0 < t_1 < \dots < t_k = n)$, define a local segment score

$$A_{ij} = M_{ij} c(i, j) h(j) \mathbb{1}_{\{j < n\}},$$

where $c(i, j)$ is a segment-cohesion term and $h(j)$ is an optional prior factor for an interior boundary at coordinate j . Treating these as separate objects prevents a physical segment-length prior from being confused with a local boundary hazard.

The posterior weight of a partition with k segments is proportional to

$$p(k) \prod_{r=1}^k A_{t_{r-1}+1, t_r}.$$

Sum-product dynamic programming sums these weights over all admissible partitions. A max-sum version replaces summation by maximization and stores backpointers to recover the joint MAP partition. After all reachable segment scores have been tabulated, the worst-case recursion cost is $\mathcal{O}(k_{\max} n^2)$.

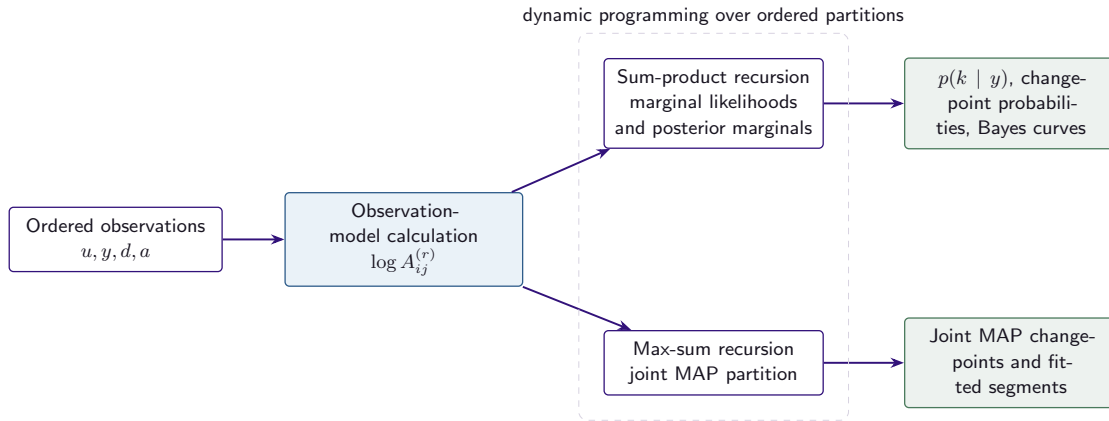


Figure 1. Statistical structure of BayesBreak. Observation-family-specific segment calculations feed two distinct dynamic-programming recursions: sum-product for posterior marginalization and max-sum with backtracking for the joint MAP partition.

2.2 Irregular designs

Irregular coordinates $x_1 < \dots < x_n$ enter through the prior on partitions rather than through an index-only approximation. Segment cohesion may depend on physical span, while a boundary hazard may depend on a local design gap. The two factors have different probability interpretations and must be specified separately. The existing archived length-aware well-log fit is a real computation, but its former Poisson-gap interpretation is not retained until the implemented prior matches that derivation.

2.3 Multiple sequences and known groups

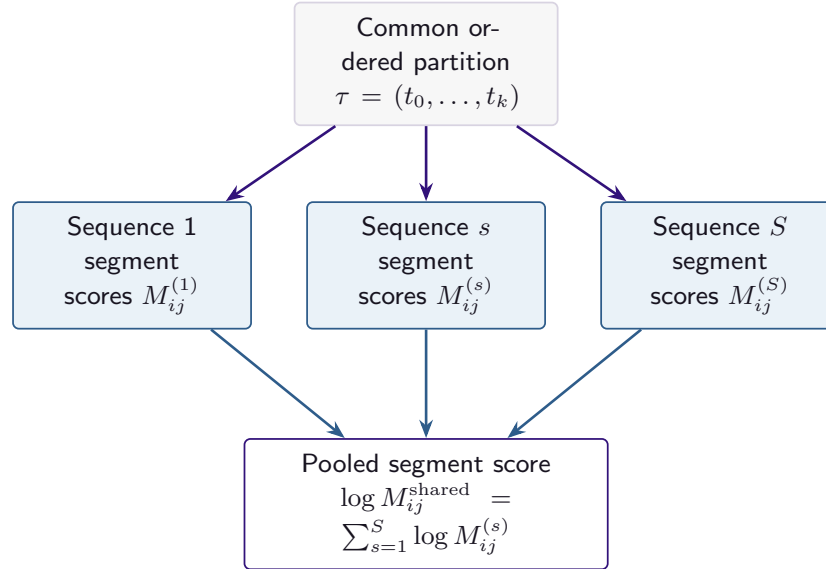
For aligned sequences $s = 1, \dots, S$ that are conditionally independent given a common partition, the segment marginal likelihood is

$$M_{ij}^{\text{shared}} = \prod_{s=1}^S M_{ij}^{(s)}.$$

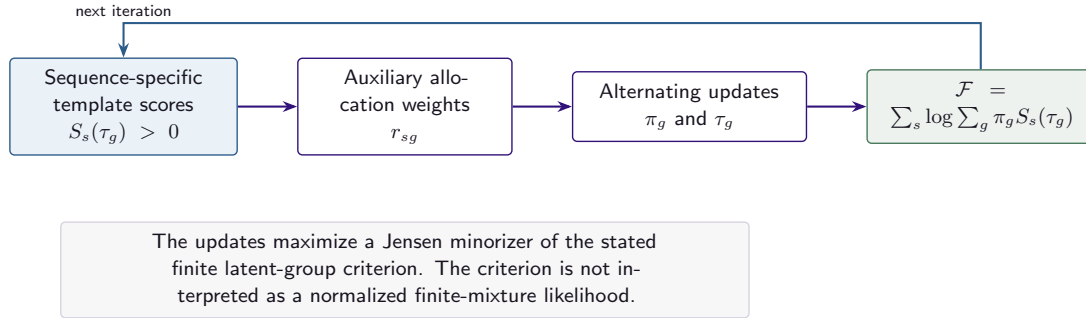
The same partition recursion can then be applied to the pooled segment scores. Known groups use the same construction within each group. The shared-boundary model is appropriate only when the candidate coordinates are aligned and a common partition is substantively plausible.

2.4 Finite latent groups

When group membership is unknown, the current method optimizes a finite criterion over group weights, sequence responsibilities, and segmentation templates. A Jensen minorization produces normalized auxiliary weights, and each template update is solved by a responsibility-weighted max-sum recursion. This is a minorization–maximization procedure for the stated criterion. Its component scores are not normalized sampling densities, so the method is not described as a conventional finite-mixture likelihood and no mixture-identifiability result is claimed.



(a) Common partition across conditionally independent sequences.



(b) Finite latent-group allocation and template updates.

Figure 2. Hierarchical extensions. The shared-boundary model pools sequence-specific segment marginal likelihoods. The latent-group model alternates responsibility and template updates for the explicitly stated finite criterion.

3 Established results and executed evidence

The current scientific status separates mathematical results, executed empirical results, implementation diagnostics, and work that remains planned. No result is strengthened merely because it appears in an executive summary.

3.1 Mathematical status

Statement	Current status	Interpretation
Conjugate exponential-family segment marginal likelihoods	Exactly derived under the stated regularity, propriety, and moment conditions	Supplies exact family-specific segment calculations from cumulative sufficient statistics.
Sum-product partition recursion	Established	Computes fixed-count marginal likelihoods, segment-count posterior probabilities, changepoint marginals, and posterior signal moments.
Max-sum recursion with backtracking	Established	Returns the joint MAP partition; it is not replaced by marginal changepoint modes.
Shared-boundary pooling	Established for aligned, conditionally independent sequences with a common partition	Subject-specific segment marginal likelihoods multiply before partition inference.
Finite-candidate shared-boundary concentration	Established under a positive average expected log-score gap and law-of-large-numbers conditions	Does not follow from one informative sequence as the number of sequences grows.
Nonconjugate error propagation	Conditional theorem	A uniform reachable-segment log-score error yields explicit bounds on partition marginal likelihoods and posterior odds.
Routine-specific nonconjugate rates	Proof obligation	No rate claim is used without method-specific assumptions and a complete proof.

Table 1. Status of the main mathematical statements.

3.2 Selected executed simulation results

Real executed numerical results

In the archived studies, the repeated Gaussian boundary-calibration experiment produced ECE approximately 0.010 and Brier score approximately 0.011. A two-group latent-allocation experiment produced 96% hard assignment accuracy at $\sigma = 1.0$ with 24 sequences. The $k_{\max} = 20$ timing study increased from 0.0455 seconds at $n = 50$ to 1.0585 seconds at $n = 800$. These are results for their stated designs, not universal performance guarantees.

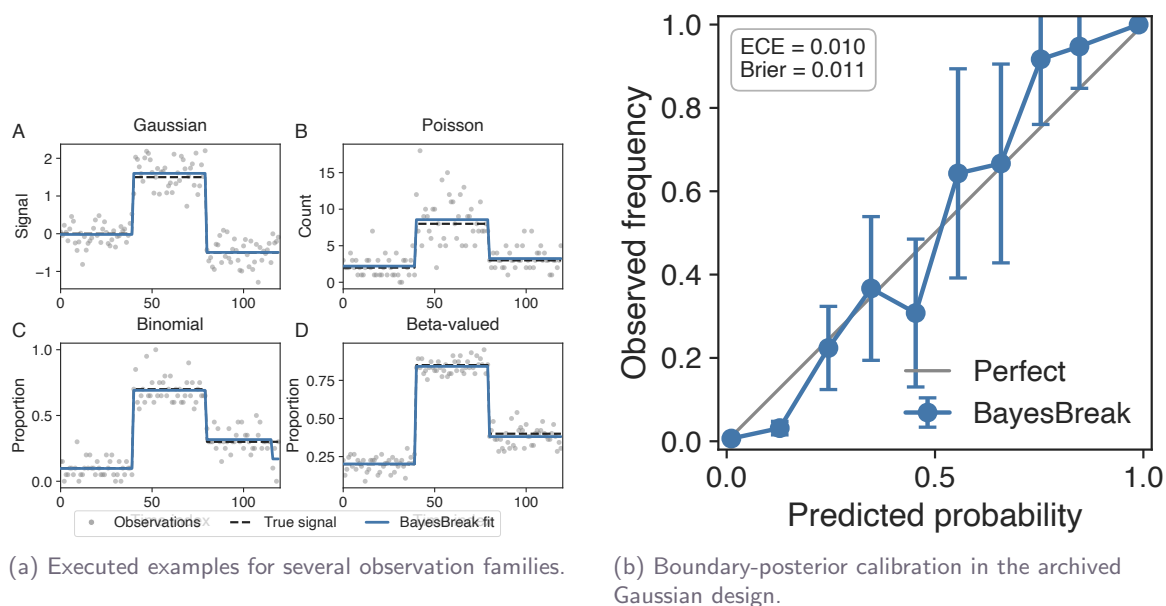


Figure 3. Examples of observation-family portability and posterior calibration. The partition recursion is shared; each panel retains the sampling model and prior appropriate to its observation family.

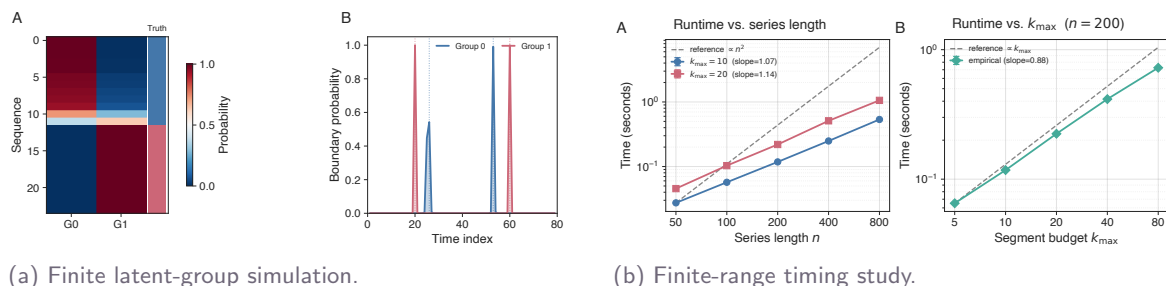


Figure 4. Executed latent-allocation and timing results. The timing slopes observed over $n = 50, \dots, 800$ are empirical summaries over that range and do not replace the worst-case $\mathcal{O}(k_{\max} n^2)$ complexity statement.

3.3 Applied studies

Four archived applications illustrate well-log, array-CGH, equity-volatility, and methylation analyses. They show that the implementation can fit heterogeneous ordered datasets using the corresponding segment models and hierarchical structures. They are descriptive case studies because independent reference changepoints are unavailable. Fitted vertical markers are MAP changepoints, not externally established geological, genomic, or macroeconomic events.

Application	Executed model	Selected archived output	Permitted interpretation
Well-log	Gaussian segmentation with two prior specifications	$\hat{k} = 23$ and 25 ; corresponding log marginal likelihoods -4989.2835 and -4997.1478	Descriptive segmentation and prior sensitivity; no external-boundary accuracy claim.
Array-CGH	Shared-boundary fit to 43 aligned sequences of length 2,215	$\hat{k} = 15$; pooled log marginal likelihood 76359.7996	Descriptive common-partition fit; not a Bayes factor against the separately factorized independent analysis.
SPX	Gaussian volatility response and Bernoulli threshold-incidence response	$\hat{k} = 29$ and 50 ; log marginal likelihoods -1296.6537 and -114.4100	Two different response models; contextual event alignment only.
Methylation	Beta-valued segmentation on the methylKit chromosome-21 example	$\hat{k} = 15$; log marginal likelihood -9518.6675 ; $n = 1904$	Descriptive segmentation; corrected family-specific prediction remains to be run.

Table 2. Selected real executed outputs and their interpretation limits. Values are rounded here for readability; archived records remain unchanged.

4 Current limitations and excluded interpretations

The principal limitations are scientific and computational, not presentational.

Two historical calculations excluded from named conclusions

Array-CGH comparator. The archived comparator calculation flattened a $43 \times 2,215$ output to 95,245 values while the BayesBreak reference boundaries remained on the 2,215-probe coordinate axis. It is a real execution of an incompatible comparison and is excluded from comparator conclusions.

Methylation held-out score. The archived value -387.50040013308154 for $m = 381$ was produced through a Gaussian fallback for Beta observations and implicit assignment of post-training coordinates to the endpoint segment. It is a real execution but not a valid Beta posterior-predictive score and is excluded from predictive conclusions.

Additional limits are as follows.

- The latent-group optimization can have label permutations, duplicate templates, local optima, or collapsed groups. One implementation path returns a stale objective value and must be repaired before restart selection relies on that value.
- The nonconjugate theorem is conditional on a uniform error bound over reachable segment log marginal likelihoods. An approximate segment routine without such a bound produces an approximate posterior whose discrepancy is not certified by the current theory.
- The independent Phase 6 bounded verification collected 179 tests: 173 passed, five were skipped because of optional dependencies, one EP logistic-normal test exceeded a 20-second per-test cap, and none failed. The remaining timeout is neither a pass nor a failure; the historical 157/5/17 state remains in the result ledger.
- The current empirical archive does not contain a fully fairness-controlled comparator benchmark, broad repeated uncertainty intervals for every setting, comprehensive misspecification grids, or independent reference changepoints for the four applications.
- Prediction outside the fitted coordinate range requires an explicit extrapolation policy; silent endpoint assignment is not acceptable.

These limitations do not change the research direction. They determine which claims are currently established, which empirical conclusions are justified, and which implementation tasks must precede a final release.

5 Implementation and validation sequence

The existing package and archived results are not replaced by a new implementation in this phase. The next implementation cycle should apply the approved statistical definitions, preserve archived outputs, and produce new result identifiers for every corrected computation.

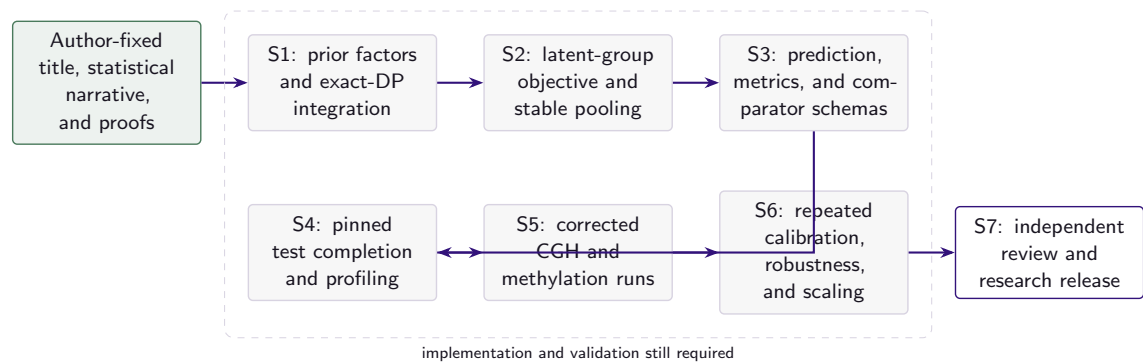


Figure 5. Approved implementation and validation sequence. Green denotes completed manuscript-level work; neutral nodes denote implementation or validation still required. A stage advances only after its stated numerical and statistical criteria are met.

Stage	Principal tasks	Required output	Advance criterion
S1	Separate segment cohesion from boundary hazard; integrate both with sum-product and max-sum recursions	Small- n exhaustive-enumeration report and prior-sensitivity fixtures	Marginal likelihoods, posterior quantities, and MAP partitions agree with enumeration under seeded cases.
S2	Repair latent-group objective reporting; harden grouped log-domain aggregation	Objective traces, restart-selection tests, high-precision aggregation checks	Every exit returns the final objective; monotonicity holds to tolerance; adversarial magnitude tests preserve rankings.
S3	Implement Beta posterior prediction, explicit extrapolation, one-to-one boundary matching, and comparator-axis validation	Family-specific prediction tests, metric fixtures, schema validation	No family fallback; out-of-range behavior is explicit; incompatible axes fail before metric computation.
S4	Pin the environment and resolve the 17 bounded-run tests	Test manifest, profiles, environment hash, termination status	Every required test passes, fails, skips, or remains blocked with an owner and reason; no timeout is counted as a pass.
S5	Recompute the invalid CGH comparator and methylation predictive analyses with new result IDs	New sidecars, parent-result links, figures, tables, and interpretation records	Correct coordinate/family definitions, schema validation, source hashes, and independent review of the resulting conclusions.
S6	Run expanded calibration, robustness, misspecification, fairness, and scaling studies	Claim-mapped reports with repetitions and uncertainty intervals	Pre-specified positive, mixed, null, and falsifying outcomes are reported without changing the primary endpoint after observing results.
S7	Complete release review	Clean builds, reproducibility record, release notes, permanent artifact locations	Statistical, numerical, software, and source review completed with all unresolved items disclosed.

Table 3. Implementation stages and acceptance criteria. Task-level details are in the synchronized coding-agent handoff.

5.1 Resource profile

The work requires four coordinated roles, which may be held by fewer people when expertise overlaps:

- a statistical-method lead to review priors, posterior prediction, approximation assumptions, metrics, and claim language;
- a scientific-software engineer to implement the task registry, schemas, tests, and result lineage;
- an empirical lead to run simulations and applications under fixed protocols and tuning budgets; and
- an independent reviewer for the final proof, numerical, comparator, and release checks.

The dominant computation is expected in repeated simulation, nonconjugate segment integration, and multi-sequence comparator studies. Exact resource estimates should be produced after the profiled test run and a small pilot of the expanded simulation grid; the current archive does not support a defensible fixed compute estimate.

6 Decision points and release criteria

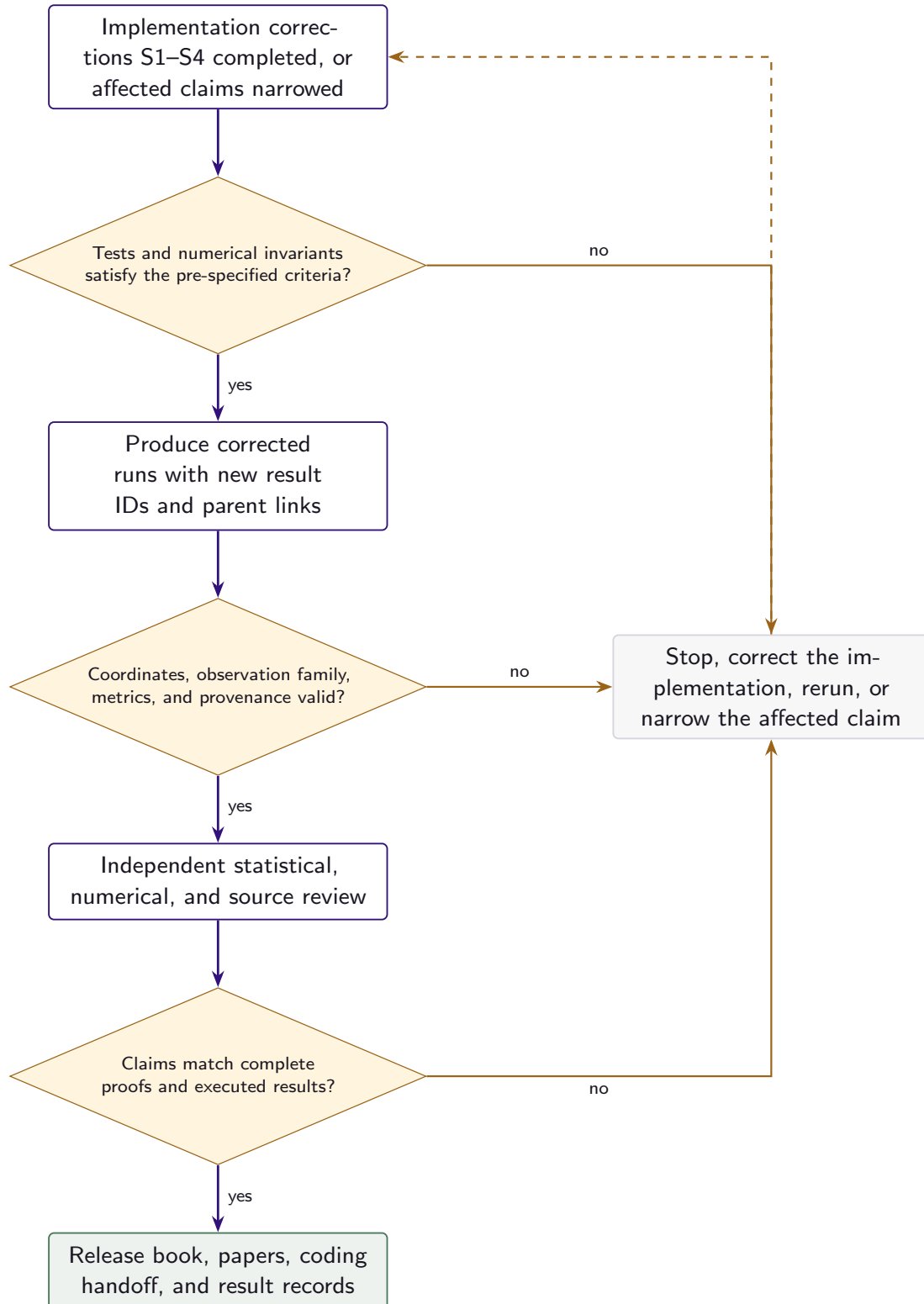


Figure 6. Decision logic for a research release. Failure of a criterion leads to correction, a narrower claim, or an explicitly incomplete release; it does not permit substitution of an unrelated method or reinterpretation of an invalid calculation.

6.1 Minimum conditions for a defensible release

A release suitable for external technical review should satisfy the following conditions.

1. The author-approved title and generalized hierarchical Bayesian segmentation narrative remain unchanged.
2. All theorem-like statements used as established results have complete proofs; the routine-specific nonconjugate rate remains labeled as a proof obligation unless proved.
3. The implementation corrections in Stages S1–S4 are complete or any incomplete item is explicitly removed from the affected claim.
4. The invalid CGH comparator and methylation predictive calculation are not used unless corrected runs with new result IDs pass their statistical and schema checks.
5. Every reported comparison states the reference set, tuning budget, coordinate axis, matching rule, repetitions, and uncertainty treatment.
6. The book, main paper, executive summary, coding handoff, and presentation handoffs agree on terminology, notation, result status, and limitations.

6.2 Decisions required from the research lead

Decision	Options	Consequence
Scope of the next implemen- tation cycle	Minimum correction set S1–S5; or S1–S7 including expanded validation	The minimum route repairs invalid calculations and unresolved software states. The expanded route supports stronger empirical assessment but requires more compute and review time.
Comparator study	Agreement-only analysis; or a fairness- controlled external benchmark with in- dependent reference changepoints where available	Agreement-only results cannot sup- port external accuracy. A bench- mark requires task-specific baselines, matched tuning budgets, and inde- pendent references.
Nonconjugate claims	Retain the conditional propagation theo- rem only; or prove method-specific rates under explicit assumptions	The first option is already mathe- matically defensible. The second is a separate theoretical project and should not delay release unless es- sential to the target venue.
Latent-group extension	Retain the present finite criterion; or de- velop a normalized probabilistic mixture as future work	A normalized mixture changes the model and requires new theory, im- plementation, and experiments. It is not a silent revision of the current method.
Release destination	Select journal venue, permanent reposi- tory, and data/artifact location	These choices determine formatting, anonymity, archival metadata, and reproducibility links but do not alter the scientific method.

Table 4. Research and release decisions that remain open.

7 Executive conclusion

BayesBreak addresses a well-defined statistical problem: Bayesian multiple-changepoint segmentation with observation-family generality and hierarchical structure for irregular designs, multiple sequences, and

observed or latent groups. Its principal mathematical mechanism is standard in form but broad in use: integrate segment parameters, combine segment marginal likelihoods with a factorized partition prior, and use dynamic programming for posterior marginalization or joint MAP optimization. The contribution lies in the generalized hierarchical formulation, the exact results under stated conditions, the explicit distinction between exact and approximate segment calculations, and the implementation across several observation families and structured settings.

The technical book and journal paper establish the current method without changing its title or research direction. The archived numerical values are real executed results and remain read-only. The next work is not a conceptual rebranding exercise. It is a defined program of statistical implementation, numerical verification, corrected computation, repeated validation, and external release review.

The immediate decision is whether to fund only the minimum corrective cycle through Stage S5 or the broader validation and release program through Stage S7. In either case, claims must remain matched to proofs, probability models, optimization objectives, and executed experiments. No invalid comparison, unsupported approximation rate, or unresolved test state should be converted into a stronger conclusion through wording alone.

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